



EUROPEAN PATENT & TECHNOLOGY

The Only Waste Product is Clean Water.

How BioTC turns municipal organic liability into high-energy **Eco-Coal** and clean local water.

The Clean Water Paradox

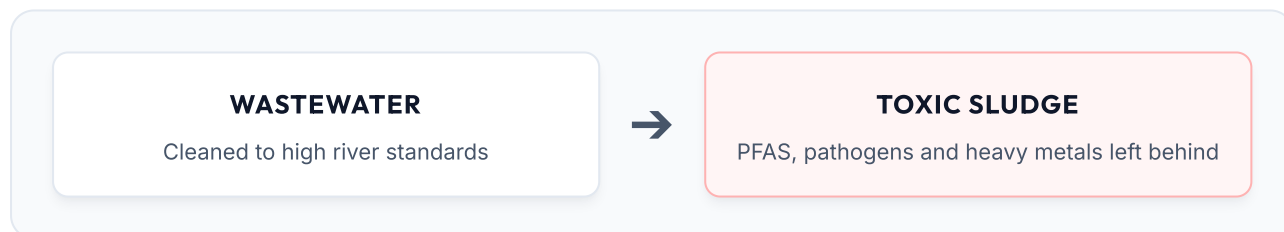
We cleaned our rivers, but we poisoned our soil.

Every wastewater treatment plant does half the job perfectly: it returns clean water to our rivers. But where does the dirt go?

The Toxic Accumulation Liability

Traditional treatment generates pathogen-ridden sewage sludge. Dumping mud in landfills or mixing it with lime simply hides the smell. It does not destroy **microplastics, pharmaceuticals, and PFAS ("forever chemicals")**. They wait in the soil, ready to seep back into our drinking water.

As EU environmental requirements tighten, water companies face skyrocketing gate fees, stricter transport regulations, and chemical bans. Landfills are filling up. The "cleaning water" loop is broken because we are leaving a toxic trail on land.



"The cleaner we make our water, the more toxic mud we pile on our land. It is time to complete the loop."

Zero Sludge. Net-Positive Energy.

We don't bury the problem. We rearrange it.

Using BioTC's advanced **Organic Composition Unit**, we take wet organic waste and put it through a fast, pressurized thermal loop. In just two hours, the molecular structure of the sludge is permanently transformed.

Our patented HTC process does not burn the waste—it pressure-cooks it, mimicking natural coal creation in minutes. The result is a complete eradication of sludge volume, turning your biggest liability into two valuable assets:



Eco-Coal

A sterile, odorless, high-energy carbon biofuel (20–25 MJ/kg, equivalent to high-grade coal). Can be used locally for heat generation or sold directly to industrial boilers.

CARBON ASSET



Clean Water

All pathogen activity, bacterial loads, microplastics, and organic chemicals (PFAS) are thermally destroyed, releasing clean water back into the municipal treatment plant cycle.

WATER ASSET

"We are not managing waste. We are manufacturing clean fuel."

Molecular Composition

Lego for Organic Molecules

Think of organic waste as a pile of disordered Lego bricks. Under high pressure and temperature, our closed-loop system acts as a master builder. It breaks down the wet organic structure, kills all pathogens, and neatly reconstructs it into dry, solid carbon energy.



The Power of the Integrated Loop

Why BioTC?

Other technologies try to solve this problem in halves. BioTC is the only patented system that integrates **Thermal Hydrolysis (TH)** and **Hydrothermal Carbonization (HTC)** into a single, energy-positive system.

Feature	Landfills & Hauling	Competitor THP (Cambi, TerraNova)	BioTC Loop PATENTED
Sludge Volume Reduction	✗ 0%	⚠ Up to 50%	✓ Up to 80%
Pathogen & PFAS Destruction	✗ None	⚠ Partial	✓ 100% Destroyed
Energy Profile	● Heavy cost	● Self-Consuming	● Net-Positive
Biogas Yield Increase	✗ 0%	⚠ +30%	✓ +60%
Financial Impact	💸 Continuous cost	⚠ Slow ROI	📈 4-6 year ROI

By recycling the high-temperature liquid stream directly into the digestion tanks, the BioTC system doubles biogas output, allowing the plant to sell electricity and heat back to the local grid.

Proven & Backed by Giants

We don't ask you to be the guinea pig.

The transition from waste risk to clean carbon has already been mapped and verified by municipalities and leading European engineers.

MUNICIPAL MILESTONE



Lubin Water Utility (MPWiK Lubin)

In October 2025, the municipal water utility in Lubin, Poland, officially signed and purchased copyrights to implement BioTheCon's integrated TH-AD-HTC layout as part of its Phase II plant modernization.

ACADEMIC ENDORSEMENT



AGH University of Krakow

The patented process was developed in close collaboration with professors of thermal and environmental engineering at AGH University, supported by official state and national research grants.

EPC CONTRACTOR



INTROL Group S.A.

All system builds, civil engineering, and integration works are performed under General Contract by INTROL, a major listed European infrastructure contractor with a track record since 1990.

The Cost of Waiting

Don't let your legacy leak away.

In a few years, environmental laws will make landfilling wet sludge illegal, and carbon taxes will skyrocket. The mayors and CEOs who wait will be forced to build expensive treatment systems using municipal funds alone.

The leaders who act today have access to green subsidies (covering up to 70% of costs) and get to declare to their community: **"We stopped poisoning our water. We did it first, and we did it right."**

Get a Free Organic Waste Audit

- 1 Send a Sample**
We will analyze your municipal or industrial sludge at our lab.
- 2 Receive Audit Report**
Get detailed yield metrics for Eco-Coal, water quality, and biogas boost.
- 3 Calculate ROI & Payback**
See the exact financial payback timeline and plant footprint requirements.

CONTACT US

 ul. Daszyńskiego 34/3, 44-100
Gliwice

 andrzej.krop@biotc.pl

 +48 608 003 458

BioTC